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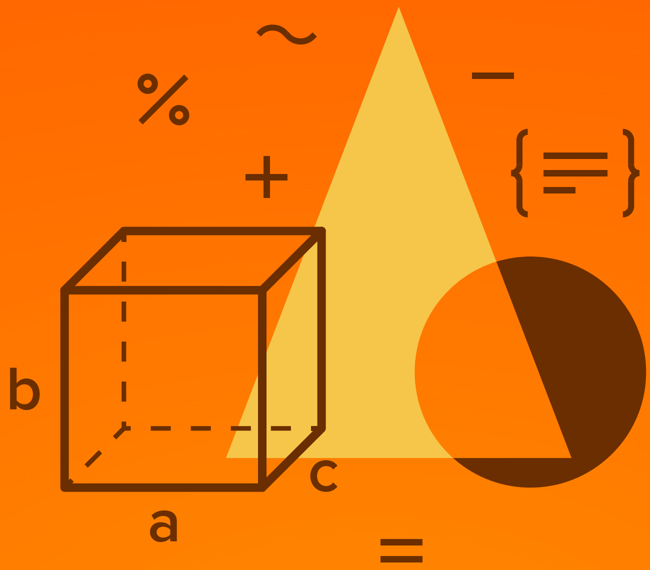


MATHEMATICS

JEE (MAIN+ADVANCED)

ENTHUSIAST COURSE

EXERCISE

Vector-3D

English Medium

EXERCISE (O-1)

Straight Objective Type

1. Four points A(+1, –1, 1) ; B(1, 3, 1) ; C(4, 3, 1) and D(4, – 1, 1) taken in order are the vertices of
(A) a parallelogram which is neither a rectangle nor a rhombus
(B) rhombus
(C) an isosceles trapezium
(D) a cyclic quadrilateral.

VT0005

2. Let α, β & γ be distinct real numbers. The points whose position vector's are $\alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$; $\beta\hat{i} + \gamma\hat{j} + \alpha\hat{k}$ and $\gamma\hat{i} + \alpha\hat{j} + \beta\hat{k}$
(A) are collinear
(B) form an equilateral triangle
(C) form a scalene triangle
(D) form a right angled triangle

VT0006

3. Let A(0, –1, 1), B(0, 0, 1), C(1, 0, 1) are the vertices of a ΔABC. If R and r denotes the circumradius and inradius of ΔABC, then $\frac{r}{R}$ has value equal to
(A) $\tan\frac{3\pi}{8}$
(B) $\cot\frac{3\pi}{8}$
(C) $\tan\frac{\pi}{12}$
(D) $\cot\frac{\pi}{12}$

VT0008

4. Consider the points A, B and C with position vectors $\left(-2\hat{i}+3\hat{j}+5\hat{k}\right)$, $\left(\hat{i}+2\hat{j}+3\hat{k}\right)$ and $7\hat{i}-\hat{k}$ respectively.
Statement-1: The vector sum, $\overrightarrow{AB}+\overrightarrow{BC}+\overrightarrow{CA} = \vec{0}$
because
Statement-2: A, B and C form the vertices of a triangle.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

VT0015

5. If the vector $6\hat{i} - 3\hat{j} - 6\hat{k}$ is decomposed into vectors parallel and perpendicular to the vector $\hat{i} + \hat{j} + \hat{k}$ then the vectors are :
(A) $-\left(\hat{i}+\hat{j}+\hat{k}\right)$ & $7\hat{i}-2\hat{j}-5\hat{k}$
(B) $-2\left(\hat{i}+\hat{j}+\hat{k}\right)$ & $8\hat{i}-\hat{j}-4\hat{k}$
(C) $+2\left(\hat{i}+\hat{j}+\hat{k}\right)$ & $4\hat{i}-5\hat{j}-8\hat{k}$
(D) none

VT0018

6. Let A(1, 2, 3), B(0, 0, 1), C(–1, 1, 1) are the vertices of a ΔABC.

(i) The equation of internal angle bisector through A to side BC is

(A) $\vec{r} = \hat{i} + 2\hat{j} + 3\hat{k} + \mu(3\hat{i} + 2\hat{j} + 3\hat{k})$
(B) $\vec{r} = \hat{i} + 2\hat{j} + 3\hat{k} + \mu(3\hat{i} + 4\hat{j} + 3\hat{k})$
(C) $\vec{r} = \hat{i} + 2\hat{j} + 3\hat{k} + \mu(3\hat{i} + 3\hat{j} + 2\hat{k})$
(D) $\vec{r} = \hat{i} + 2\hat{j} + 3\hat{k} + \mu(3\hat{i} + 3\hat{j} + 4\hat{k})$

(ii) The equation of median through C to side AB is

(A) $\vec{r} = -\hat{i} + \hat{j} + \hat{k} + p(3\hat{i} - 2\hat{k})$
(B) $\vec{r} = -\hat{i} + \hat{j} + \hat{k} + p(3\hat{i} + 2\hat{k})$
(C) $\vec{r} = -\hat{i} + \hat{j} + \hat{k} + p(-3\hat{i} + 2\hat{k})$
(D) $\vec{r} = -\hat{i} + \hat{j} + \hat{k} + p(3\hat{i} + 2\hat{j})$

(iii) The area (ΔABC) is equal to

(A) $\frac{9}{2}$ (B) $\frac{\sqrt{17}}{2}$ (C) $\frac{17}{2}$ (D) $\frac{7}{2}$

VT0020

7. A, B, C & D are four points in a plane with pv's $\vec{a}, \vec{b}, \vec{c}$ & $\vec{d}$ respectively such that $(\vec{a} - \vec{d}) \cdot (\vec{b} - \vec{c}) = (\vec{b} - \vec{d}) \cdot (\vec{c} - \vec{a}) = 0$. Then for the triangle ABC, D is its

(A) incentre (B) circumcentre (C) orthocentre (D) centroid

VT0026

8. $\vec{a}$ and $\vec{b}$ are unit vectors inclined to each other at an angle α, $\alpha \in (0, \pi)$ and $|\vec{a} + \vec{b}| < 1$. Then $\alpha \in$

(A) $\left(\frac{\pi}{3}, \frac{2\pi}{3}\right)$ (B) $\left(\frac{2\pi}{3}, \pi\right)$ (C) $\left(0, \frac{\pi}{3}\right)$ (D) $\left(\frac{\pi}{4}, \frac{3\pi}{4}\right)$

VT0027

9. Let $\hat{a}, \hat{b}, \hat{c}$ are three unit vectors such that $\hat{a} + \hat{b} + \hat{c}$ is also a unit vector. If pairwise angles between $\hat{a}, \hat{b}, \hat{c}$ are θ_1, θ_2 and θ_3 respectively then $\cos\theta_1 + \cos\theta_2 + \cos\theta_3$ equals

(A) 3 (B) – 3 (C) 1 (D) – 1

VT0029

10. A tangent is drawn to the curve $y = \frac{8}{x^2}$ at a point $A(x_1, y_1)$, where $x_1 = 2$. The tangent cuts the x-axis at point B. Then the scalar product of the vectors $\overrightarrow{AB}$ & $\overrightarrow{OB}$ is

(A) 3 (B) – 3 (C) 6 (D) – 6

VT0030

11. Cosine of an angle between the vectors $(\vec{a}+\vec{b})$ and $(\vec{a}-\vec{b})$ if $|\vec{a}| = 2$, $|\vec{b}| = 1$ and $\vec{a} \wedge \vec{b} = 60°$ is

(A) $\sqrt{3/7}$ (B) $9/\sqrt{21}$ (C) $3/\sqrt{7}$ (D) none

VT0031

12. The vector equations of two lines L_1 and L_2 are respectively

$\vec{r} = 17\hat{i} - 9\hat{j} + 9\hat{k} + \lambda(3\hat{i} + \hat{j} + 5\hat{k})$ and $\vec{r} = 15\hat{i} - 8\hat{j} - \hat{k} + \mu(4\hat{i} + 3\hat{j})$

I L_1 and L_2 are skew lines

II $(11, -11, -1)$ is the point of intersection of L_1 and L_2

III $(-11, 11, 1)$ is the point of intersection of L_1 and L_2

IV $\cos^{-1}\left(3/\sqrt{35}\right)$ is the acute angle between L_1 and L_2

then , which of the following is true?

(A) II and IV (B) I and IV (C) IV only (D) III and IV

VT0034

13. For some non zero vector $\vec{V}$, if the sum of $\vec{V}$ and the vector obtained from $\vec{V}$ by rotating it by an angle 2α equals to the vector obtained from $\vec{V}$ by rotating it by α then the value of α, is

(A) $2n\pi \pm \frac{\pi}{3}$ (B) $n\pi \pm \frac{\pi}{3}$ (C) $2n\pi \pm \frac{2\pi}{3}$ (D) $n\pi \pm \frac{2\pi}{3}$

where n is an integer.

VT0036

14. Let $\vec{u}, \vec{v}, \vec{w}$ be such that $|\vec{u}| = 1$, $|\vec{v}| = 2$, $|\vec{w}| = 3$. If the projection of $\vec{v}$ along $\vec{u}$ is equal to that of $\vec{w}$ along $\vec{u}$ and vectors $\vec{v}$, $\vec{w}$ are perpendicular to each other then $|\vec{u} - \vec{v} + \vec{w}|$ equals

(A) 2 (B) $\sqrt{7}$ (C) $\sqrt{14}$ (D) 14

VT0037

15. If $\vec{a}$ and $\vec{b}$ are non zero, non collinear, and the linear combination

$(2x - y)\vec{a} + 4\vec{b} = 5\vec{a} + (x - 2y)\vec{b}$ holds for real x and y then x + y has the value equal to

(A) – 3 (B) 1 (C) 17 (D) 3

VT0038

16. Given an equilateral triangle ABC with side length equal to 'a'. Let M and N be two points respectively on the side AB and AC such that $\overrightarrow{AN} = K\overrightarrow{AC}$ and $\overrightarrow{AM} = \frac{\overrightarrow{AB}}{3}$. If $\overrightarrow{BN}$ and $\overrightarrow{CM}$ are orthogonal then the value of K is equal to

(A) $\frac{1}{5}$ (B) $\frac{1}{4}$ (C) $\frac{1}{3}$ (D) $\frac{1}{2}$

VT0039

17. If $\vec{p}$ & $\vec{s}$ are not perpendicular to each other and $\vec{r} \times \vec{p} = \vec{q} \times \vec{p}$ & $\vec{r} \cdot \vec{s} = 0$, then $\vec{r} =$

(A) $\vec{p} \cdot \vec{s}$
(B) $\vec{q} + \left(\frac{\vec{q} \cdot \vec{p}}{\vec{p} \cdot \vec{s}}\right)\vec{p}$
(C) $\vec{q} - \left(\frac{\vec{q} \cdot \vec{s}}{\vec{p} \cdot \vec{s}}\right)\vec{p}$
(D) $\vec{q} + \mu\vec{p}$ for all scalars μ

VT0040

18. If $\vec{u}$ and $\vec{v}$ are two vectors such that $|\vec{u}| = 3$; $|\vec{v}| = 2$ and $|\vec{u} \times \vec{v}| = 6$ then the correct statement is

(A) $\vec{u} \wedge \vec{v} \in (0, 90°)$
(B) $\vec{u} \wedge \vec{v} \in (90°, 180°)$
(C) $\vec{u} \wedge \vec{v} = 90°$
(D) $(\vec{u} \times \vec{v}) \times \vec{u} = 6\vec{v}$

VT0041

19. Given a parallelogram OACB. The lengths of the vectors $\overrightarrow{OA}$, $\overrightarrow{OB}$ & $\overrightarrow{AB}$ are a, b & c respectively. The scalar product of the vectors $\overrightarrow{OC}$ & $\overrightarrow{OB}$ is :

(A) $\frac{a^2 - 3b^2 + c^2}{2}$ (B) $\frac{3a^2 + b^2 - c^2}{2}$ (C) $\frac{3a^2 - b^2 + c^2}{2}$ (D) $\frac{a^2 + 3b^2 - c^2}{2}$

VT0042

20. Vectors $\vec{a}$ & $\vec{b}$ make an angle $\theta = \frac{2\pi}{3}$. If $|\vec{a}| = 1$, $|\vec{b}| = 2$ then $\left\{\left(\vec{a} + 3\vec{b}\right) \times \left(3\vec{a} - \vec{b}\right)\right\}^2 =$

(A) 225 (B) 250 (C) 275 (D) 300

VT0043

21. If the vector product of a constant vector $\overrightarrow{OA}$ with a variable vector $\overrightarrow{OB}$ in a fixed plane OAB be a constant vector, then locus of B is :

(A) a straight line perpendicular to $\overrightarrow{OA}$

(B) a circle with centre O radius equal to $\left|\overrightarrow{OA}\right|$

(C) a straight line parallel to $\overrightarrow{OA}$

(D) none of these

VT0044

22. The vectors $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$; $\vec{b} = 2\hat{i} - \hat{j} + \hat{k}$ & $\vec{c} = 3\hat{i} + \hat{j} + 4\hat{k}$ are so placed that the end point of one vector is the starting point of the next vector. Then the vectors are -

(A) not coplanar

(B) coplanar but cannot form a triangle

(C) coplanar but can form a triangle

(D) coplanar & can form a right angled triangle

VT0046

23. Given the vectors

$$\vec{u} = 2\hat{i} - \hat{j} - \hat{k}$$

$$\vec{v} = \hat{i} - \hat{j} + 2\hat{k}$$

$$\vec{w} = \hat{i} - \hat{k}$$

If the volume of the parallelopiped having $-c\vec{u}$, $\vec{v}$ and $c\vec{w}$ as concurrent edges, is 8 then 'c' can be equal to

(A) ± 2 (B) 4 (C) 8 (D) can not be determined

VT0047

24. Given $\bar{a} = x\hat{i} + y\hat{j} + 2\hat{k}$, $\bar{b} = \hat{i} - \hat{j} + \hat{k}$, $\bar{c} = \hat{i} + 2\hat{j}$; $(\bar{a} \,\hat{}\, \bar{b}) = \pi/2, \bar{a} \cdot \bar{c} = 4$ then

(A) $[\bar{a}\ \bar{b}\ \bar{c}]^2 = |\bar{a}|$ (B) $[\bar{a}\ \bar{b}\ \bar{c}] = |\bar{a}|$

(C) $[\bar{a}\ \bar{b}\ \bar{c}] = 0$ (D) $[\bar{a}\ \bar{b}\ \bar{c}] = |\bar{a}|^2$

VT0048

25. Let $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$; $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$; $\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$ be three non-zero vectors such that $\vec{c}$ is a unit vector perpendicular to both $\vec{a}$ & $\vec{b}$. If the angle between $\vec{a}$ & $\vec{b}$ is $\frac{\pi}{6}$, then $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}^2 =$

(A) 0

(B) 1

(C) $\frac{1}{4}(a_1^2 + a_2^2 + a_3^2)(b_1^2 + b_2^2 + b_3^2)$

(D) $\frac{3}{4}(a_1^2 + a_2^2 + a_3^2)(b_1^2 + b_2^2 + b_3^2)(c_1^2 + c_2^2 + c_3^2)$

VT0049

26. A rigid body rotates with constant angular velocity ω about the line whose vector equation is, $\vec{r} = \lambda\left(\hat{i} + 2\hat{j} + 2\hat{k}\right)$. The speed of the particle at the instant it passes through the point with p.v. $2\hat{i} + 3\hat{j} + 5\hat{k}$ is :

(A) $\omega\sqrt{2}$ (B) 2ω (C) $\omega/\sqrt{2}$ (D) none

VT0052

27. Given 3 vectors $\vec{V}_1 = a\hat{i} + b\hat{j} + c\hat{k}$; $\vec{V}_2 = b\hat{i} + c\hat{j} + a\hat{k}$; $\vec{V}_3 = c\hat{i} + a\hat{j} + b\hat{k}$

In which one of the following conditions $\vec{V}_1, \vec{V}_2$ and $\vec{V}_3$ are linearly independent?

(A) $a + b + c = 0$ and $a^2 + b^2 + c^2 \neq ab + bc + ca$

(B) $a + b + c = 0$ and $a^2 + b^2 + c^2 = ab + bc + ca$

(C) $a + b + c \neq 0$ and $a^2 + b^2 + c^2 = ab + bc + ca$

(D) $a + b + c \neq 0$ and $a^2 + b^2 + c^2 \neq ab + bc + ca$

VT0053

28. Given unit vectors $\vec{m}$, $\vec{n}$ & $\vec{p}$ such that angle between $\vec{m}$ & $\vec{n}$ = angle between $\vec{p}$ and $\left(\vec{m} \times \vec{n}\right) = \pi/6$, then $\left[\vec{n}\ \vec{p}\ \vec{m}\right] =$

(A) $\sqrt{3}/4$ (B) 3/4 (C) 1/4 (D) none

VT0054

29. The altitude of a parallelopiped whose three coterminous edges are the vectors, $\vec{A} = \hat{i} + \hat{j} + \hat{k}$; $\vec{B} = 2\hat{i} + 4\hat{j} - \hat{k}$ & $\vec{C} = \hat{i} + \hat{j} + 3\hat{k}$ with $\vec{A}$ and $\vec{B}$ as the sides of the base of the parallelopiped, is

(A) $2/\sqrt{19}$ (B) $4/\sqrt{19}$ (C) $2\sqrt{38}/19$ (D) none

VT0056

30. Consider Δ ABC with A ≡ $(\vec{a})$; B ≡ $(\vec{b})$ & C ≡ $(\vec{c})$. If $\vec{b}.(\vec{a} + \vec{c}) = \vec{b}.\vec{b} + \vec{a}.\vec{c}$; $\left|\vec{b} - \vec{a}\right| = 3$; $\left|\vec{c} - \vec{b}\right| = 4$ then the angle between the medians $\vec{AM}$ & $\vec{BD}$ is

(A) $\pi - \cos^{-1}\left(\frac{1}{5\sqrt{13}}\right)$ (B) $\pi - \cos^{-1}\left(\frac{1}{13\sqrt{5}}\right)$

(C) $\cos^{-1}\left(\frac{1}{5\sqrt{13}}\right)$ (D) $\cos^{-1}\left(\frac{1}{13\sqrt{5}}\right)$

VT0057

EXERCISE (O-2)

Multiple Correct Answer Type

1. If $\vec{a}$, $\vec{b}$, $\vec{c}$ be three non zero vectors satisfying the condition $\vec{a} \times \vec{b} = \vec{c}$ & $\vec{b} \times \vec{c} = \vec{a}$ then which of the following always hold(s) good?

(A) $\vec{a}$, $\vec{b}$, $\vec{c}$ are orthogonal in pairs
(B) $\left[\vec{a}\ \vec{b}\ \vec{c}\right] = \left|\vec{b}\right|$
(C) $\left[\vec{a}\ \vec{b}\ \vec{c}\right] = \left|\vec{c}\right|^2$
(D) $\left|\vec{b}\right| = \left|\vec{c}\right|$

VT0074

2. Given the following information about the non zero vectors $\vec{A}, \vec{B}$ and $\vec{C}$

(i) $(\vec{A} \times \vec{B}) \times \vec{A} = \vec{0}$ (ii) $\vec{B} \cdot \vec{B} = 4$ (iii) $\vec{A} \cdot \vec{B} = -6$ (iv) $\vec{B} \cdot \vec{C} = 6$

Which one of the following holds good?

(A) $\vec{A} \times \vec{B} = \vec{0}$ (B) $\vec{A} \cdot (\vec{B} \times \vec{C}) = 0$ (C) $\vec{A} \cdot \vec{A} = 8$ (D) $\vec{A} \cdot \vec{C} = -9$

VT0075

3. If $\vec{A}, \vec{B}, \vec{C}$ and $\vec{D}$ are four non zero vectors in the same plane no two of which are collinear then which of the following hold(s) good?

(A) $(\vec{A} \times \vec{B}) \cdot (\vec{C} \times \vec{D}) = 0$
(B) $(\vec{A} \times \vec{C}) \cdot (\vec{B} \times \vec{D}) \neq 0$
(C) $(\vec{A} \times \vec{B}) \times (\vec{C} \times \vec{D}) = \vec{0}$
(D) $(\vec{A} \times \vec{C}) \times (\vec{B} \times \vec{D}) \neq \vec{0}$

VT0076

4. If $\vec{a}$, $\vec{b}$, $\vec{c}$ & $\vec{d}$ are the pv's of the points A, B, C & D respectively in three dimensional space & satisfy the relation $3\vec{a} - 2\vec{b} + \vec{c} - 2\vec{d} = \vec{0}$, then :

(A) A, B, C & D are coplanar
(B) the line joining the points B & D divides the line joining the point A & C in the ratio 2 : 1.
(C) the line joining the points A & C divides the line joining the points B & D in the ratio 1 : 1
(D) the four vectors $\vec{a}$, $\vec{b}$, $\vec{c}$ & $\vec{d}$ are linearly dependent.

VT0077

5. The vectors $\vec{u} = \begin{bmatrix} 6 \\ -3 \\ 2 \end{bmatrix}$; $\vec{v} = \begin{bmatrix} 2 \\ 6 \\ 3 \end{bmatrix}$; $\vec{w} = \begin{bmatrix} 3 \\ 2 \\ -6 \end{bmatrix}$

(A) form a left handed system
(B) form a right handed system
(C) are linearly independent
(D) are such that each is perpendicular to the plane containing the other two.

VT0078

6. If $\vec{a}$, $\vec{b}$, $\vec{c}$ are non-zero, non-collinear vectors such that a vector

$\vec{p} = ab\cos\left(2\pi - \left(\vec{a} \wedge \vec{b}\right)\right)\ \vec{c}$ and a vector $\vec{q} = ac\cos\left(\pi - \left(\vec{a} \wedge \vec{c}\right)\right)\ \vec{b}$ then $\vec{p} + \vec{q}$ is

(A) parallel to $\vec{a}$ (B) perpendicular to $\vec{a}$

(C) coplanar with $\vec{b}$ & $\vec{c}$ (D) coplanar with $\vec{a}$ and $\vec{c}$

VT0079

7. Which of the following statement(s) hold good ?

(A) if $\vec{a}\cdot\vec{b} = \vec{a}\cdot\vec{c} \Rightarrow \vec{b} = \vec{c} \quad (\vec{a} \neq \vec{0})$

(B) if $\vec{a}\times\vec{b} = \vec{a}\times\vec{c} \Rightarrow \vec{b} = \vec{c} \quad (\vec{a} \neq \vec{0})$

(C) if $\vec{a}\cdot\vec{b} = \vec{a}\cdot\vec{c}$ and $\vec{a}\times\vec{b} = \vec{a}\times\vec{c} \Rightarrow \vec{b} = \vec{c} \quad (\vec{a} \neq 0)$

(D) if $\vec{v}_1, \vec{v}_2, \vec{v}_3$ are non coplanar vectors and $\vec{k}_1 = \dfrac{\vec{v}_2\times\vec{v}_3}{\vec{v}_1\cdot(\vec{v}_2\times\vec{v}_3)}$; $\vec{k}_2 = \dfrac{\vec{v}_3\times\vec{v}_1}{\vec{v}_1\cdot(\vec{v}_2\times\vec{v}_3)}$

and $\vec{k}_3 = \dfrac{\vec{v}_1\times\vec{v}_2}{\vec{v}_1\cdot(\vec{v}_2\times\vec{v}_3)}$ then $\vec{k}_1\cdot(\vec{k}_2\times\vec{k}_3) = \dfrac{1}{\vec{v}_1\cdot(\vec{v}_2\times\vec{v}_3)}$

VT0080

8. If the line $\vec{r} = 2\hat{i} - \hat{j} + 3\hat{k} + \lambda(\hat{i} + \hat{j} + \sqrt{2}\,\hat{k})$ makes angles α, β, γ with xy, yz and zx planes respectively then which of the following are not possible?

(A) $\sin^2\alpha + \sin^2\beta + \sin^2\gamma = 2$ & $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$

(B) $\tan^2\alpha + \tan^2\beta + \tan^2\gamma = 7$ & $\cot^2\alpha + \cot^2\beta + \cot^2\gamma = 5/3$

(C) $\sin^2\alpha + \sin^2\beta + \sin^2\gamma = 1$ & $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 2$

(D) $\sec^2\alpha + \sec^2\beta + \sec^2\gamma = 10$ & $\mathrm{cosec}^2\alpha + \mathrm{cosec}^2\beta + \mathrm{cosec}^2\gamma = 14/3$

VT0081

9. If a, b, c are different real numbers and $a\hat{i} + b\hat{j} + c\hat{k}$; $b\hat{i} + c\hat{j} + a\hat{k}$ & $c\hat{i} + a\hat{j} + b\hat{k}$ are position vectors of three non-collinear points A, B & C then :

(A) centroid of triangle ABC is $\dfrac{a+b+c}{3}\left(\hat{i} + \hat{j} + \hat{k}\right)$

(B) $\hat{i} + \hat{j} + \hat{k}$ is equally inclined to the three vectors

(C) perpendicular from the origin to the plane of triangle ABC meet at centroid

(D) triangle ABC is an equilateral triangle.

VT0082

10. A vector of magnitude 10 along the normal to the curve $3x^2 + 8xy + 2y^2 - 3 = 0$ at its point $P(1, 0)$ can be

(A) $6\hat{i} + 8\hat{j}$ (B) $-6\hat{i} + 8\hat{j}$ (C) $6\hat{i} - 8\hat{j}$ (D) $-6\hat{i} - 8\hat{j}$

VT0083

11. Let OAB be a regular triangle with side length unity (O being the origin). Also M,N are the points of trisection of AB,M being closer to A and N closer to B. Position vectors of A,B,M and N are $\vec{a}, \vec{b}, \vec{m}$ and $\vec{n}$ respectively. Which of the following hold(s) good ?

(A) $\vec{m} = x\vec{a} + y\vec{b} \Rightarrow \frac{2}{3}$ and $y = \frac{1}{3}$ (B) $\vec{m} = x\vec{a} + y\vec{b} \Rightarrow x = \frac{5}{6}$ and $y = \frac{1}{6}$

(C) $\vec{m}.\vec{n}$ equals $\frac{13}{18}$ (D) $\vec{m}.\vec{n}$ equals $\frac{15}{18}$

VT0084

12. Which of the following statement(s) is(are) incorrect ?

(A) The relation $|(\vec{u} \times \vec{v})| = |\vec{u}.\vec{v}|$ is only possible if atleast one of the vectors $\vec{u}$ and $\vec{v}$ is null vector.

(B) Every vector contained in the line $\vec{r}(t) = \langle 1 + 2t, 1 + 3t, 1 + 4t \rangle$ is parallel to the vector $\langle 1,1,1 \rangle$.

(C) If scalar triple product of three vectors, $\vec{u}, \vec{v}, \vec{w}$ is larger than $|\vec{u} \times \vec{v}|$ then $|\vec{w}| > 1$.

(D) The distance between the x-axis and the line $x = y = 1$ is $\sqrt{2}$.

VT0086

13. Given three vectors $\vec{U} = 2\hat{i} + 3\hat{j} - 6\hat{k}$; $\vec{V} = 6\hat{i} + 2\hat{j} + 3\hat{k}$; $\vec{W} = 3\hat{i} - 6\hat{j} - 2\hat{k}$

Which of the following hold good for the vectors $\vec{U}$, $\vec{V}$ and $\vec{W}$?

(A) $\vec{U}$, $\vec{V}$ and $\vec{W}$ are linearly dependent (B) $(\vec{U} \times \vec{V}) \times \vec{W} = \vec{0}$

(C) $\vec{U}$, $\vec{V}$ and $\vec{W}$ form a triplet of mutually perpendicular vectors (D) $\vec{U} \times (\vec{V} \times \vec{W}) = \vec{0}$

VT0087

14. Which of the following statement(s) is/are true in respect of the lines

$\vec{r} = \vec{a} + \lambda\vec{b}$; $\vec{r} = \vec{c} + \mu\vec{d}$ where $\vec{b} \times \vec{d} \neq 0$

(A) acute angle between the lines is $\cos^{-1}\left(\frac{|\vec{b}.\vec{d}|}{|\vec{b}||\vec{d}|}\right)$

(B) The lines would intersect if $[\vec{c}\ \vec{b}\ \vec{d}] = [\vec{a}\ \vec{b}\ \vec{d}]$

(C) The lines will be skew if $[\vec{c} - \vec{a}\ \vec{b}\ \vec{d}] \neq 0$

(D) If the lines intersect at $\vec{r} = \vec{r}_0$, then the equation of the plane containing the lines is $[\vec{r} - \vec{r}_0\ \vec{b}\ \vec{d}] = 0$

VT0088

15. Let $\vec{a}$ and $\vec{b}$ be two non-zero and non-collinear vectors then which of the following is/are always correct ?

(A) $\vec{a}\times\vec{b}=[\vec{a}\ \vec{b}\ \hat{i}]\hat{i}+[\vec{a}\ \vec{b}\ \hat{j}]\hat{j}+[\vec{a}\ \vec{b}\ \hat{k}]\hat{k}$

(B) $\vec{a}.\vec{b}=(\vec{a}.\hat{i})(\vec{b}.\hat{i})+(\vec{a}.\hat{j}).(\vec{b}.\hat{j})+(\vec{a}.\hat{k})(\vec{b}.\hat{k})$

(C) if $\vec{u}=\hat{a}-(\hat{a}.\hat{b})\hat{b}$ and $\vec{v}=\hat{a}\times\hat{b}$ then $|\vec{u}|=|\vec{v}|$

(D) if $\vec{c}=\vec{a}\times(\vec{a}\times\vec{b})$ and $\vec{d}=\vec{b}\times(\vec{a}\times\vec{b})$ then $\vec{c}+\vec{d}=\vec{0}$

VT0089

16. The value(s) of $\alpha\in[0,2\pi]$ for which vector $\vec{a}=\hat{i}+3\hat{j}+(\sin 2\alpha)\hat{k}$ makes an obtuse angle with the z-axis and the vectors $\vec{b}=(\tan\alpha)\hat{i}-\hat{j}+2\sqrt{\sin\frac{\alpha}{2}}\,\hat{k}$ and $\vec{c}=(\tan\alpha)\hat{i}+(\tan\alpha)\hat{j}-3\sqrt{\operatorname{cosec}\frac{\alpha}{2}}\,\hat{k}$ are orthogonal, is/are

(A) $\tan^{-1}3$ (B) $\pi-\tan^{-1}2$ (C) $\pi+\tan^{-1}3$ (D) $2\pi-\tan^{-1}2$

VT0192

17. The vector $\hat{i}+x\hat{j}+3\hat{k}$ is rotated through an angle of $\cos^{-1}\frac{11}{14}$ and doubled in magnitude, then it becomes $4\hat{i}+(4x-2)\hat{j}+2\hat{k}$. The value of $'x'$ **CANNOT** be :

(A) $-\frac{2}{3}$ (B) $\frac{2}{3}$ (C) $-\frac{20}{17}$ (D) 2

VT0193

18. The vector $\vec{c}$, parallel to the internal bisector of the angle between the vectors $\vec{a}=7\hat{i}-4\hat{j}-4\hat{k}$ and $\vec{b}=-2\hat{i}-\hat{j}+2\hat{k}$ with $|\vec{c}|=5\sqrt{6}$, is :

(A) $\frac{5}{3}\left(\hat{i}-7\hat{j}+2\hat{k}\right)$ (B) $\frac{5}{3}\left(\hat{i}+7\hat{j}-2\hat{k}\right)$ (C) $\frac{5}{3}\left(\hat{i}+7\hat{j}-2\hat{k}\right)$ (D) $\frac{5}{3}\left(-\hat{i}-7\hat{j}+2\hat{k}\right)$

VT0194

19. A line passes through a point A with position vector $3\hat{i}+\hat{j}-\hat{k}$ and is parallel to the vector $2\hat{i}-\hat{j}+2\hat{k}$. If P is a point on this line such that AP = 15 units, then the position vector of the point P is/are

(A) $13\hat{i}+4\hat{j}-9\hat{k}$ (B) $13\hat{i}-4\hat{j}+9\hat{k}$

(C) $7\hat{i}-6\hat{j}+11\hat{k}$ (D) $-7\hat{i}+6\hat{j}-11\hat{k}$

VT0195

20. Which of the followings is/are correct :

(A) The angle between the two straight lines $\vec{r} = 3\hat{i} - 2\hat{j} + 4\hat{k} + \lambda\left(-2\hat{i} + \hat{j} + 2\hat{k}\right)$ and $\vec{r} = \hat{i} + 3\hat{j} - 2\hat{k} + \mu\left(3\hat{i} - 2\hat{j} + 6\hat{k}\right)$ is $\cos^{-1}\left(\frac{4}{21}\right)$

(B) $\left(\vec{r}.\hat{i}\right)\left(\hat{i} \times \vec{r}\right) + \left(\vec{r}.\hat{j}\right)\left(\hat{j} \times \vec{r}\right) + \left(\vec{r}.\hat{k}\right)\left(\hat{k} \times \vec{r}\right) = \vec{0}$

(C) The force determined by the vector $\vec{r} = (1, -8, -7)$ is resolved along three mutually perpendicular directions, one of which is in the direction of the vector $\vec{a} = 2\hat{i} + 2\hat{j} + \hat{k}$. Then the vector component of the force $\vec{r}$ in the direction of the vector $\vec{a}$ is $-\frac{7}{3}\left(2\hat{i} + 2\hat{j} + \hat{k}\right)$

(D) The cosine of the angle between any two diagonals of a cube is $\frac{1}{3}$. **VT0196**

21. If the distance between points $(\alpha, 5\alpha, 10\alpha)$ from the point of intersection of the line. $\vec{r} = \left(2\hat{i} - \hat{j} + 2\hat{k}\right) + \lambda\left(2\hat{i} + 4\hat{j} + 12\hat{k}\right)$ and plane $\vec{r}.\left(\hat{i} - \hat{j} + \hat{k}\right) = 5$ is 13 units, then value of α may be

(A) 1 (B) –1 (C) 4 (D) $\frac{80}{63}$

VT0197

22. $\hat{a}$ and $\hat{b}$ are two given unit vectors at right angle. The unit vector equally inclined with $\hat{a}$, $\hat{b}$ and $\hat{a} \times \hat{b}$ will be :

(A) $-\frac{1}{\sqrt{3}}\left(\hat{a} + \hat{b} + \hat{a} \times \hat{b}\right)$ (B) $\frac{1}{\sqrt{3}}\left(\hat{a} + \hat{b} + \hat{a} \times \hat{b}\right)$

(C) $\frac{1}{\sqrt{3}}\left(\hat{a} + \hat{b} - \hat{a} \times \hat{b}\right)$ (D) $-\frac{1}{\sqrt{3}}\left(\hat{a} + \hat{b} - \hat{a} \times \hat{b}\right)$ **VT0198**

23. If $\vec{a}$, $\vec{b}$, $\vec{c}$ and $\vec{d}$ are unit vectors such that $(\vec{a} \times \vec{b}).(\vec{c} \times \vec{d}) = \lambda$ and $\vec{a}.\vec{c} = \frac{\sqrt{3}}{2}$, then

(A) $\vec{a}, \vec{b}, \vec{c}$ are coplanar if $\lambda = 1$

(B) Angle between $\vec{b}$ and $\vec{d}$ is 30° if $\lambda = -1$

(C) angle between $\vec{b}$ and $\vec{d}$ is 150° if $\lambda = -1$

(D) If $\lambda = 1$ then angle between $\vec{b}$ and $\vec{c}$ is 60° **VT0199**

24. Let $\bar{p}, \bar{q}, \bar{r}$ are non zero vectors and $\bar{\alpha} = \bar{p} \times (\bar{q} \times \bar{r})$ and $\bar{\beta} = (\bar{p} \times \bar{q}) \times \bar{r}$. If $\bar{\alpha} = \bar{\beta}$ then which of the following holds goods

(A) $\bar{p}$ and $\bar{q}$ are orthogonal.

(B) $\bar{p}$ and $\bar{r}$ are collinear

(C) $\bar{q}$ and $\bar{r}$ are orthogonal

(D) $\bar{q} = \lambda(\bar{p} \times \bar{r})$ where 'λ' is scalar **VT0200**

25. If $|\vec{a}| = |\vec{b}| = |\vec{c}| = |\vec{d}| = 1$ such that $(\vec{a} \times \vec{b}).(\vec{c} \times \vec{d}) = 1$ and $\vec{a} \cdot \vec{c} = \frac{1}{2}$, then which of the following can be correct ?

(A) $\vec{a} \cdot \vec{b} = 0$

(B) $\vec{a} \cdot (\vec{b} \times \vec{c}) = 0$

(C) $\vec{a} \cdot (\vec{b} \times \vec{c}) \neq 0$

(D) $|\vec{a} + \vec{b} + \vec{c}| = \sqrt{4 + \sqrt{3}}$ **VT0201**

EXERCISE (O-3)

Linked Comprehension Type

Paragraph for Question 1 to 2

Vectors are essential tools for undrstanding and solving problems in physics, engineering and mathematics. One key aspect of working with vectors is finding the angle between them, which can provide valuable information about the relationship between the two vectors. The dot product, also known as the scalar product or inner product, is a crucual operation that allows us to determine the angle between two vectors.

The dot product of two vectors $\vec{A}$ and $\vec{B}$ is defined as :

$\vec{A}\cdot\vec{B} = |\vec{A}||\vec{B}|\cos(\theta)$

where $\vec{A}$ and $\vec{B}$ represent the magnitudes of the vectors A and B, respectively and θ is the angle between the vectors.

From the definition above, we can easily deduce the formula to find the angle between the two vectors :

$\theta = \arccos\left[(\vec{A}\cdot\vec{B})/(|\vec{A}||\vec{B}|)\right]$

1. In a quadrilateral ABCD, $\overrightarrow{AC}$ is the bisector of the $\left(\overrightarrow{AB}\ \hat{}\ \overrightarrow{AD}\right)$ which is $\frac{2\pi}{3}$,

$15\left|\overrightarrow{AC}\right| = 3\left|\overrightarrow{AB}\right| = 5\left|\overrightarrow{AD}\right|$ then $\cos\left(\overrightarrow{BA}\ \hat{}\ \overrightarrow{CD}\right)$ is

(A) $-\frac{\sqrt{14}}{7\sqrt{2}}$ (B) $-\frac{\sqrt{21}}{7\sqrt{3}}$ (C) $\frac{2}{\sqrt{7}}$ (D) $\frac{2\sqrt{7}}{14}$

VT0071

2. If the two adjacent sides of two rectangles are represented by the vectors $\vec{p} = 5\vec{a} - 3\vec{b}\,;\vec{q} = -\vec{a} - 2\vec{b}$ and $\vec{r} = -4\vec{a} - \vec{b}\,;\vec{s} = -\vec{a} + \vec{b}$ respectively, then the angle between the vectors $\vec{x} = \frac{1}{3}(\vec{p} + \vec{r} + \vec{s})$ and $\vec{y} = \frac{1}{5}(\vec{r} + \vec{s})$

(A) is $-\cos^{-1}\left(\frac{19}{5\sqrt{43}}\right)$

(B) is $\cos^{-1}\left(\frac{19}{5\sqrt{43}}\right)$

(C) is $\pi - \cos^{-1}\left(\frac{19}{5\sqrt{43}}\right)$

(D) cannot be evaluated

VT0072

Paragraph for Question 3 to 5

Consider three vectors $\vec{p} = \hat{i} + \hat{j} + \hat{k}$, $\vec{q} = 2\hat{i} + 4\hat{j} - \hat{k}$ and $\vec{r} = \hat{i} + \hat{j} + 3\hat{k}$ and let $\vec{s}$ be a unit vector, then

3. $\vec{p}$, $\vec{q}$ and $\vec{r}$ are

(A) linearly dependent

(B) can form the sides of a possible triangle

(C) such that the vectors $(\vec{q} - \vec{r})$ is orthogonal to $\vec{p}$

(D) such that each one of these can be expressed as a linear combination of the other two

VT0090

4. If $(\vec{p}\times\vec{q})\times\vec{r} = u\vec{p}+v\vec{q}+w\vec{r}$, then $(u+v+w)$ equals to

(A) 8 (B) 2 (C) – 2 (D) 4

VT0090

5. the magnitude of the vector $(\vec{p}\cdot\vec{s})(\vec{q}\times\vec{r}) + (\vec{q}\cdot\vec{s})(\vec{r}\times\vec{p}) + (\vec{r}\cdot\vec{s})(\vec{p}\times\vec{q})$ is

(A) 4 (B) 8 (C) 18 (D) 2

VT0090

Paragraph for Question 6 to 8

The quadratic equation in y, $(a-\sin x-\sin^2 x)y^2+(b-\cos x-\cos^2 x)y+c=0$ is satisfied by each value of λ for which vectors, $\vec{v}_1=\hat{i}+\lambda\hat{j},\ \vec{v}_2=\lambda\hat{i}+\hat{j}+\hat{k}$ & $\vec{v}_3=\hat{i}+\hat{j}+\lambda\hat{k}$ are linearly dependent. Let $\lambda_0\in Q$ be one of the values of λ.

On the basis of above information, answer the following questions :

6. Maximum value of a + b + c is - **VT0202**

7. For $\lambda=\lambda_0$ value of $|(\vec{v}_1\times\vec{v}_2)\times\vec{v}_3|$ is - **VT0203**

8. Number of values of λ for which $\vec{v}_1,\vec{v}_2$ & $\vec{v}_3$ form a triangle - **VT0204**

Matrix Match Type

9.

	List-I		List-II
(I)	P is point in the plane of the triangle ABC. pv's of A,B and C are $\vec{a},\vec{b}$ and $\vec{c}$ respectively with respect to P as the origin. If $(\vec{b}+\vec{c}).(\vec{b}-\vec{c})=0$ and $(\vec{c}+\vec{a}).(\vec{c}-\vec{a})=0$, then w.r.t. the triangle ABC,P is its **VT0091**	(P)	centroid
(II)	If $\vec{a},\vec{b},\vec{c}$ are the position vectors of the three non collinear points A,B and C respectively such that the vector $\vec{V}=\overrightarrow{PA}+\overrightarrow{PB}+\overrightarrow{PC}$ is a null vector then w.r.t. the ΔABC, P is its **VT0092**	(Q)	orthocentre
(III)	If P is a point inside the ΔABC such that the vector $\vec{R}=(BC)\overrightarrow{PA}+(CA)(\overrightarrow{PB})+(AB)(\overrightarrow{PC})$ is a null vector then w.r.t. the ΔABC, P is its **VT0093**	(R)	Incentre
(IV)	If P is a point in the plane of the triangle ABC such that the scalar product $\overrightarrow{PA}.\overrightarrow{CB}$ and $\overrightarrow{PB}.\overrightarrow{AC}$ vanishes, then w.r.t. the ΔABC, P is its **VT0094**	(S)	circumcentre

(A) I → P; II → Q; III → R; IV → S
(B) I → S; II → P; III → R; IV → Q
(C) I → R; II → P; III → Q; IV → S
(D) I → R; II → S; III → Q; IV → P

10. **Column-I** **Column-II**

	Column-I		Column-II
(A)	If $\vec{a}, \vec{b}, \vec{c}$ are non coplanar unit vectors, such that $\vec{a} \times (\vec{b} \times \vec{c}) = \frac{\vec{b}+\vec{c}}{\sqrt{2}}$, then the angle between $\vec{a}$ and $\vec{b}$ lies in	(P)	$\left(\frac{\pi}{4}, \frac{\pi}{2}\right)$
(B)	Four vectors $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ are such that $(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = \vec{0}$. Let P_1 and P_2 be planes determined by the pair of vectors $\vec{a}, \vec{b}$ and $\vec{c}$, $\vec{d}$ respectively, then angle between P_1 and P_2 lies in	(Q)	$\left(-\frac{\pi}{4}, \frac{3\pi}{4}\right)$
		(R)	$\left(\frac{\pi}{6}, \frac{5\pi}{6}\right)$
(C)	If $\vec{a}$ and $\vec{b}$ are two unit vectors such that $2\vec{a} - \vec{b}$ and $4\vec{a} + 5\vec{b}$ are perpendicular to each other, then angle between $\vec{a}$ and $\vec{b}$ lies in		
(D)	If $\|\vec{a}\| = 3$, $\|\vec{b}\| = 5$, $\|\vec{c}\| = 7$ and $\vec{a} + \vec{b} + \vec{c} = \vec{0}$, then angle between $\vec{a}$ and $\vec{b}$ lies in	(S)	$\left(-\frac{\pi}{6}, \frac{7\pi}{12}\right)$

VT0205

EXERCISE (O-4)

Numerical Grid Type

1. A rigid body rotates about an axis through the origin with an angular velocity $10\sqrt{3}$ radians/sec. If $\vec{\omega}$ points in the direction of $\hat{i}+\hat{j}+\hat{k}$ and the equation to the locus of the points having tangential speed 20 m/sec. is $x^2+y^2+z^2-axy-byz-czx-2=0$, then $(a+b+c)$ is equal to **VT0073**

2. Let $\vec{a},\vec{b},\vec{c}$ be three non-zero vectors which are pairwise non-collinear. If $\vec{a}+3\vec{b}$ is collinear with $\vec{c}$ and $\vec{b}+2\vec{c}$ is colliner with $\vec{a}$, then $|\vec{a}+3\vec{b}+6\vec{c}|$ is : **VT0147**

3. If $\vec{a}$ and $\vec{b}$ are vectors in space given by $\vec{a}=\dfrac{\hat{i}-2\hat{j}}{\sqrt{5}}$ and $\vec{b}=\dfrac{2\hat{i}+\hat{j}+3\hat{k}}{\sqrt{14}}$, then the value of $\left(2\vec{a}+\vec{b}\right).\left[\left(\vec{a}\times\vec{b}\right)\times\left(\vec{a}-2\vec{b}\right)\right]$ is **VT0169**

4. If $\vec{a}$, $\vec{b}$ and $\vec{c}$ are unit vectors satisfying $|\vec{a}-\vec{b}|^2+|\vec{b}-\vec{c}|^2+|\vec{c}-\vec{a}|^2=9$, then $|2\vec{a}+5\vec{b}+5\vec{c}|$ is **VT0173**

5. Let $|\vec{a}|=2, |\vec{b}|=1$ and $|\vec{c}|=3$ such that $\vec{a}\wedge\vec{b}=\dfrac{\pi}{3}$, $\vec{b}\wedge\vec{c}=\dfrac{\pi}{4}$ and $\vec{c}\wedge\vec{a}=\dfrac{\pi}{2}$. If $\left|\vec{a}-2\vec{b}+\vec{c}\right|$ can be expressed as $\sqrt{p+q\sqrt{2}}$ (where $p,q\in I$), then $(p+q)$ is **VT0206**

6. If $\vec{p}=3\vec{a}-5\vec{b}$; $\vec{q}=2\vec{a}+\vec{b}$; $\vec{r}=\vec{a}+4\vec{b}$; $\vec{s}=-\vec{a}+\vec{b}$ are four vectors such that $\sin\left(\vec{p}\ \wedge\ \vec{q}\right)=1$ and $\sin\left(\vec{r}\ \wedge\ \vec{s}\right)=1$ then $\sqrt{43}\cos\left(\vec{a}\ \wedge\ \vec{b}\right)$ is : **VT0070**

7. Let $\hat{a}$ and $\hat{b}$ be two unit vectors. If the vectors $\vec{c}=\hat{a}+2\hat{b}$ and $\vec{d}=5\hat{a}-4\hat{b}$ are perpendicular to each other, then the angle between $\hat{a}$ and $\hat{b}$ is : **VT0148**

8. Two adjacent sides of a parallelogram ABCD are given by $\overrightarrow{AB}=2\hat{i}+10\hat{j}+11\hat{k}$ and $\overrightarrow{AD}=-\hat{i}+2\hat{j}+2\hat{k}$. The side AD is rotated by an acute angle α in the plane of the parallelogram so that AD becomes AD'. If AD' makes a right angle with the side AB, then the cosine of the angle α is given by - **VT0168**

9. Let $\vec{r}=\sin x\left(\vec{a}\times\vec{b}\right)+\cos y\left(\vec{b}\times\vec{c}\right)+2\left(\vec{c}\times\vec{a}\right)$, where $\vec{a},\vec{b}$ & $\vec{c}$ are non-zero and non-coplanar vectors. If $\vec{r}$ is orthogonal to $\vec{a}+\vec{b}+\vec{c}$, then minimum value of $\dfrac{4}{\pi^2}\left(x^2+y^2\right)$ is **VT0207**

10. If three points $\left(2\vec{p}-\vec{q}+3\vec{r}\right),\left(\vec{p}-2\vec{q}+\alpha\vec{r}\right)$ and $\left(\beta\vec{p}-5\vec{q}\right)$ (where $\vec{p},\vec{q},\vec{r}$ are non-coplanar vectors) are collinear, then the value of $\dfrac{1}{\alpha+\beta}$ is **VT0208**

EXERCISE (JM)

1. If the vectors $\overline{AB} = 3\hat{i} + 4\hat{k}$ and $\overline{AC} = 5\hat{i} - 2\hat{j} + 4\hat{k}$ are the sides of a triangle ABC, then the length of the median through A is : **[JEE-MAINS 2013]**

(1) $\sqrt{18}$ (2) $\sqrt{72}$ (3) $\sqrt{33}$ (4) $\sqrt{45}$

VT0150

2. Let $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}, \vec{b} = \hat{i} + 2\hat{j} - \hat{k}$ and $\vec{c} = \hat{i} + \hat{j} - 2\hat{k}$ be three vectors. A vectors of the type $\vec{b} + \lambda\vec{c}$ for some scalar λ, whose projection on $\vec{a}$ is of magnitude $\sqrt{\frac{2}{3}}$, is : **[JEE-MAINS Online 2013]**

(1) $2\hat{i} + 3\hat{j} - 3\hat{k}$ (2) $2\hat{i} + \hat{j} + 5\hat{k}$ (3) $2\hat{i} - \hat{j} + 5\hat{k}$ (4) $2\hat{i} + 3\hat{j} + 3\hat{k}$

VT0151

3. Let $\vec{a} = 2\hat{i} + \hat{j} - 2\hat{k}$, $\vec{b} = \hat{i} + \hat{j}$. If $\vec{c}$ is a vector such that $\vec{a} \bullet \vec{c} = |\vec{c}|$, $|\vec{c} - \vec{a}| = 2\sqrt{2}$ and the angle between $\vec{a} \times \vec{b}$ and $\vec{c}$ is 30º, then $\left|\left(\vec{a} \times \vec{b}\right) \times \vec{c}\right|$ equals : **[JEE-MAINS Online 2013]**

(1) $\frac{3}{2}$ (2) 3 (3) $\frac{1}{2}$ (4) $\frac{3\sqrt{3}}{2}$

VT0152

4. If $\left[\vec{a} \times \vec{b}\ \ \vec{b} \times \vec{c}\ \ \vec{c} \times \vec{a}\right] = \lambda\left[\vec{a}\,\vec{b}\,\vec{c}\right]^2$ then λ is equal to : **[JEE(Main)-2014]**

(1) 2 (2) 3 (3) 0 (4) 1

VT0153

5. Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be three non-zero vectors such that no two of them are collinear and $\left(\vec{a} \times \vec{b}\right) \times \vec{c} = \frac{1}{3}|\vec{b}|\,|\vec{c}|\,\vec{a}$. If θ is the angle between vectors $\vec{b}$ and $\vec{c}$, then a value of $\sin\theta$ is :

[JEE(Main)-2015]

(1) $\frac{2}{3}$ (2) $\frac{-2\sqrt{3}}{3}$ (3) $\frac{2\sqrt{2}}{3}$ (4) $\frac{-\sqrt{2}}{3}$

VT0154

6. Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three unit vectors such that $\vec{a} \times \left(\vec{b} \times \vec{c}\right) = \frac{\sqrt{3}}{2}\left(\vec{b} + \vec{c}\right)$. If $\vec{b}$ is not parallel to $\vec{c}$, then the angle between $\vec{a}$ and $\vec{b}$ is :- **[JEE(Main)-2016]**

(1) $\frac{5\pi}{6}$ (2) $\frac{3\pi}{4}$ (3) $\frac{\pi}{2}$ (4) $\frac{2\pi}{3}$

VT0155

7. Let $\vec{a} = 2\hat{i} + \hat{j} - 2\hat{k}$ and $\vec{b} = \hat{i} + \hat{j}$. Let $\vec{c}$ be a vector such that $|\vec{c} - \vec{a}| = 3$, $\left|(\vec{a} \times \vec{b}) \times \vec{c}\right| = 3$ and the angle between $\vec{c}$ and $\vec{a} \times \vec{b}$ be 30º. Then $\vec{a} \cdot \vec{c}$ is equal to : **[JEE(Main)-2017]**

(1) $\frac{1}{8}$ (2) $\frac{25}{8}$ (3) 2 (4) 5

VT0156

8. Let $\vec{u}$ be a vector coplanar with the vectors $\vec{a} = 2\hat{i} + 3\hat{j} - \hat{k}$ and $\vec{b} = \hat{j} + \hat{k}$. If $\vec{u}$ is perpendicular to $\vec{a}$ and $\vec{u}.\vec{b} = 24$, then $|\vec{u}|^2$ is equal to - **[JEE(Main)-2018]**

(1) 315 (2) 256 (3) 84 (4) 336

VT0157

9. Let $\vec{a} = \hat{i} - \hat{j}, \vec{b} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{c}$ be a vector such that $\vec{a} \times \vec{c} + \vec{b} = \vec{0}$ and $\vec{a}.\vec{c} = 4$, then $|\vec{c}|^2$ is equal to :- **[JEE(Main) 19]**

(1) $\frac{19}{2}$ (2) 8 (3) $\frac{17}{2}$ (4) 9

VT0158

10. Let $\sqrt{3}\hat{i} + \hat{j}, \hat{i} + \sqrt{3}\hat{j}$ and $\beta\hat{i} + (1 - \beta)\hat{j}$ respectively be the position vectors of the points A, B and C with respect to the origin O. If the distance of C from the bisector of the acute angle between OA and OB is $\frac{3}{\sqrt{2}}$, then the sum of all possible values of β is :- **[JEE(Main)-19]**

(1) 2 (2) 1 (3) 3 (4) 4

VT0159

11. Let $\vec{\alpha} = 3\hat{i} + \hat{j}$ and $\vec{\beta} = 2\hat{i} - \hat{j} + 3\hat{k}$. If $\vec{\beta} = \vec{\beta}_1 - \vec{\beta}_2$, where $\vec{\beta}_1$ is parallel to $\vec{\alpha}$ and $\vec{\beta}_2$ is perpendicular to $\vec{\alpha}$, then $\vec{\beta}_1 \times \vec{\beta}_2$ is equal to **[JEE(Main)-19]**

(1) $-3\hat{i} + 9\hat{j} + 5\hat{k}$ (2) $3\hat{i} - 9\hat{j} - 5\hat{k}$ (3) $\frac{1}{2}\left(-3\hat{i} + 9\hat{j} + 5\hat{k}\right)$ (4) $\frac{1}{2}\left(3\hat{i} - 9\hat{j} + 5\hat{k}\right)$

VT0160

12. The distance of the point having position vector $-\hat{i} + 2\hat{j} + 6\hat{k}$ from the straight line passing through the point (2, 3, –4) and parallel to the vector, $6\hat{i} + 3\hat{j} - 4\hat{k}$ is : **[JEE(Main)-19]**

(1) 7 (2) $4\sqrt{3}$ (3) $2\sqrt{13}$ (4) 6

VT0161

13. Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be three units vectors such that $\vec{a}+\vec{b}+\vec{c}=\vec{0}$. If l = $\vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a}$ and $\vec{d}=\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}$, then the ordered pair, $(\lambda,\vec{d})$ is equal to : **[JEE(Main)-20]**

(1) $\left(-\frac{3}{2},3\vec{a}\times\vec{b}\right)$　(2) $\left(-\frac{3}{2},3\vec{c}\times\vec{b}\right)$　(3) $\left(\frac{3}{2},3\vec{b}\times\vec{c}\right)$　(4) $\left(\frac{3}{2},3\vec{a}\times\vec{c}\right)$

VT0162

14. Let $\vec{a}=\hat{i}-2\hat{j}+\hat{k}$ and $\vec{b}=\hat{i}-\hat{j}+\hat{k}$ be two vectors. If $\vec{c}$ is a vector such that $\vec{b}\times\vec{c}=\vec{b}\times\vec{a}$ and $\vec{c}\cdot\vec{a}=0$, then $\vec{c}\cdot\vec{b}$ is equal to **[JEE(Main)-20]**

(1) $\frac{1}{2}$　(2) -1　(3) $-\frac{1}{2}$　(4) $-\frac{3}{2}$

VT0163

15. Let the volume of a parallelopiped whose coterminous edges are given by $\vec{u}=\hat{i}+\hat{j}+\lambda\hat{k},\vec{v}=\hat{i}+\hat{j}+3\hat{k}$ and $\vec{w}=2\hat{i}+\hat{j}+\hat{k}$ be 1 cu. unit. If q be the angle between the edges $\vec{u}$ and $\vec{w}$, then cosq can be **[JEE(Main)-20]**

(1) $\frac{7}{6\sqrt{3}}$　(2) $\frac{5}{7}$　(3) $\frac{7}{6\sqrt{6}}$　(4) $\frac{5}{3\sqrt{3}}$

VT0164

16. Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be three vectors such that $|\vec{a}|=\sqrt{3}$, $|\vec{b}|=5$, $\vec{b}\cdot\vec{c}=10$ and the angle between $\vec{b}$ and $\vec{c}$ is $\frac{\pi}{3}$. If $\vec{a}$ is perpendicular to the vector $\vec{b}\times\vec{c}$, then $\left|\vec{a}\times(\vec{b}\times\vec{c})\right|$ is equal to _____. **[JEE(Main)-20]**

VT0165

17. If the vectors, $\vec{p}=(a+1)\hat{i}+a\hat{j}+a\hat{k}$, **[JEE(Main)-20]**

$\vec{q}=a\hat{i}+(a+1)\hat{j}+a\hat{k}$ and

$\vec{r}=a\hat{i}+a\hat{j}+(a+1)\hat{k}$ $(a\in R)$ are coplanar and $3(\vec{p}.\vec{q})^2-\lambda|\vec{r}\times\vec{q}|^2=0$, then the value of l is ———.

VT0166

18. The projection of the line segment joining the points (1, –1, 3) and (2, –4, 11) on the line joining the points (–1, 2, 3) and (3, –2, 10) is ———. **[JEE(Main)-20]**

VT0167

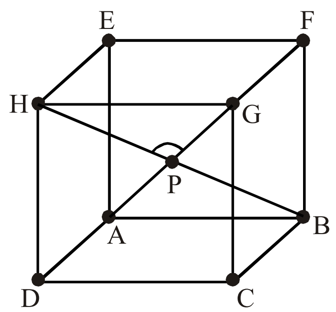
19. Let O be the origin. Let $\overline{OP}=x\hat{i}+y\hat{j}-\hat{k}$ and $\overline{OQ}=-\hat{i}+2\hat{j}+3x\hat{k}$, $x, y\in R, x>0$, be such that $|\overline{PQ}|=\sqrt{20}$ and the vector $\overline{OP}$ is perpendicular to $\overline{OQ}$. If $\overline{OR}=3\hat{i}+z\hat{j}-7\hat{k}$, $z\in R$, is coplanar with $\overline{OP}$ and $\overline{OQ}$, then the value of $x^2+y^2+z^2$ is equal to **[JEE(Main)-21]**

(1) 7　(2) 9　(3) 2　(4) 1

VT0209

20. A hall has a square floor of dimension 10m × 10m (see the figure) and vertical walls. If the angle GPH between the diagonals AG and BH is $\cos^{-1}\frac{1}{5}$ then the height of the hall (in meters) is :

[JEE(Main)-21]



(1) 5 (2) $2\sqrt{10}$ (3) $5\sqrt{3}$ (4) $5\sqrt{2}$

VT0210

21. Let $\vec{a}=\hat{i}+\hat{j}+2\hat{k}$ and $\vec{b}=-\hat{i}+2\hat{j}+3\hat{k}$. Then the vector product $(\vec{a}+\vec{b})\times\left(\left(\vec{a}\times\left((\vec{a}-\vec{b})\times\vec{b}\right)\right)\times\vec{b}\right)$ is equal to : **[JEE(Main)-21]**

(1) $5(34\hat{i}-5\hat{j}+3\hat{k})$ (2) $7(34\hat{i}-5\hat{j}+3\hat{k})$ (3) $7(30\hat{i}-5\hat{j}+7\hat{k})$ (4) $5(30\hat{i}-5\hat{j}+7\hat{k})$

VT0211

22. Let $\vec{a}, \vec{b}, \vec{c}$ be three non-coplanar vectors such that $\vec{a}\times\vec{b}=4\vec{c}$, $\vec{b}\times\vec{c}=9\vec{a}$ and $\vec{c}\times\vec{a}=\alpha\vec{b}$, $\alpha > 0$. If $|\vec{a}|+|\vec{b}|+|\vec{c}|=\frac{1}{36}$, then α is equal to ______. **[JEE(Main)-22]**

VT0212

23. Let the vectors $\vec{a}=(1+t)\hat{i}+(1-t)\hat{j}+\hat{k}$, $\vec{b}=(1-t)\hat{i}+(1+t)\hat{j}+2\hat{k}$ and $\vec{c}=t\hat{i}-t\hat{j}+\hat{k}$, $t\in R$ be such that for $\alpha,\beta,\gamma\in R$, $\alpha\vec{a}+\beta\vec{b}+\gamma\vec{c}=\vec{0} \Rightarrow \alpha=\beta=\gamma=0$. Then, the set of all values of t is :

[JEE(Main)-22]

(A) a non-empty finite set (B) equal to N

(C) equal to R – {0} (D) equal to R **VT0213**

24. Let $\hat{a}$ and $\hat{b}$ be two unit vectors such that the angle between them is $\frac{\pi}{4}$. If θ is the angle between the vectors $(\hat{a}+\hat{b})$ and $(\hat{a}+2\hat{b}+2(\hat{a}\times\hat{b}))$, then the value of $164\cos^2\theta$ is equal to :

[JEE(Main)-22]

(A) $90+27\sqrt{2}$ (B) $45+18\sqrt{2}$ (C) $90+3\sqrt{2}$ (D) $54+90\sqrt{2}$

VT0214

EXERCISE (JA)

1. Let $\overrightarrow{PR} = 3\hat{i} + \hat{j} - 2\hat{k}$ and $\overrightarrow{SQ} = \hat{i} - 3\hat{j} - 4\hat{k}$ determine diagonals of a parallelogram PQRS and $\overrightarrow{PT} = \hat{i} + 2\hat{j} + 3\hat{k}$ be another vector. Then the volume of the parallelepiped determined by the vectors $\overrightarrow{PT}$, $\overrightarrow{PQ}$ and $\overrightarrow{PS}$ is **[JEE-Advanced 2013, 2M]**

(A) 5 (B) 20 (C) 10 (D) 30

VT0175

2. Consider the set of eight vectors $V = \left\{a\hat{i} + b\hat{j} + c\hat{k} : a, b, c \in \{-1, 1\}\right\}$. Three non-coplanar vectors can be chosen from V in 2^p ways. Then p is **[JEE-Advanced 2013, 4, (–1)]**

VT0176

3. Match List-I with List-II and select the correct answer using the code given below the lists.

List-I		List-II	
P.	Volume of parallelepiped determined by vectors $\vec{a}, \vec{b}$ and $\vec{c}$ is 2. Then the volume of the parallelepiped determined by vectors $2\left(\vec{a} \times \vec{b}\right), 3\left(\vec{b} \times \vec{c}\right)$ and $\left(\vec{c} \times \vec{a}\right)$ is	1.	100
Q.	Volume of parallelepiped determined by vectors $\vec{a}, \vec{b}$ and $\vec{c}$ is 5. Then the volume of the parallelepiped determined by vectors $3\left(\vec{a} + \vec{b}\right), \left(\vec{b} + \vec{c}\right)$ and $2\left(\vec{c} + \vec{a}\right)$ is	2.	30
R.	Area of a triangle with adjacent sides determined by vectors $\vec{a}$ and $\vec{b}$ is 20. Then the area of the triangle with adjacent sides determined by vectors $\left(2\vec{a} + 3\vec{b}\right)$ and $\left(\vec{a} - \vec{b}\right)$ is	3.	24
S.	Area of a parallelogram with adjacent sides determined by vectors $\vec{a}$ and $\vec{b}$ is 30. Then the area of the parallelogram with adjacent sides determined by vectors $\left(\vec{a} + \vec{b}\right)$ and $\vec{a}$ is	4.	60

Codes :

	P	Q	R	S
(A)	4	2	3	1
(B)	2	3	1	4
(C)	3	4	1	2
(D)	1	4	3	2

[JEE-Advanced 2013, 3, (–1)]

VT0177

4. Let $\vec{x}, \vec{y}$ and $\vec{z}$ be three vectors each of magnitude $\sqrt{2}$ and the angle between each pair of them is $\frac{\pi}{3}$. If $\vec{a}$ is a nonzero vector perpendicular to $\vec{x}$ and $\vec{y}\times\vec{z}$ and $\vec{b}$ is nonzero vector perpendicular to $\vec{y}$ and $\vec{z}\times\vec{x}$, then **[JEE(Advanced)-2014, 3]**

(A) $\vec{b}=(\vec{b}.\vec{z})(\vec{z}-\vec{x})$ (B) $\vec{a}=(\vec{a}.\vec{y})(\vec{y}-\vec{z})$ (C) $\vec{a}.\vec{b}=-(\vec{a}.\vec{y})(\vec{b}.\vec{z})$ (D) $\vec{a}=(\vec{a}.\vec{y})(\vec{z}-\vec{y})$

VT0178

5. Let $\vec{a}, \vec{b}$, and $\vec{c}$ be three non-coplanar unit vectors such that the angle between every pair of them is $\frac{\pi}{3}$. If $\vec{a}\times\vec{b}+\vec{b}\times\vec{c}=p\vec{a}+q\vec{b}+r\vec{c}$, where p,q and r are scalars, then the value of $\frac{p^2+2q^2+r^2}{q^2}$ is

[JEE(Advanced)-2014, 3]
VT0179

6. Let ΔPQR be a triangle. Let $\vec{a}=\overrightarrow{QR}, \vec{b}=\overrightarrow{RP}$ and $\vec{c}=\overrightarrow{PQ}$. If $|\vec{a}|=12, |\vec{b}|=4\sqrt{3}$ and $\vec{b}.\vec{c}=24$, then which of the following is (are) true ? **[JEE 2015, 4M, –2M]**

(A) $\frac{|\vec{c}|^2}{2}-|\vec{a}|=12$ (B) $\frac{|\vec{c}|^2}{2}+|\vec{a}|=30$ (C) $|\vec{a}\times\vec{b}+\vec{c}\times\vec{a}|=48\sqrt{3}$ (D) $\vec{a}.\vec{b}=-72$

VT0180

7. Suppose that $\vec{p}, \vec{q}$ and $\vec{r}$ are three non-coplanar vectors in $\mathbb{R}^3$. Let the components of a vector $\vec{s}$ along $\vec{p}, \vec{q}$ and $\vec{r}$ be 4,3 and 5, respectively. If the components of this vector $\vec{s}$ along $\left(-\vec{p}+\vec{q}+\vec{r}\right)$, $\left(\vec{p}-\vec{q}+\vec{r}\right)$ and $\left(-\vec{p}-\vec{q}+\vec{r}\right)$ are x,y and z, respectively, then the value of 2x + y + z is

[JEE 2015, 4M, –0M]
VT0181

8. Let $\hat{u}=u_1\hat{i}+u_2\hat{j}+u_3\hat{k}$ be a unit vector in $\mathbb{R}^3$ and $\hat{w}=\frac{1}{\sqrt{6}}(\hat{i}+\hat{j}+2\hat{k})$. Given that there exists a vector $\vec{v}$ in $\mathbb{R}^3$ such that $\left|\hat{u}\times\vec{v}\right|=1$ and $\hat{w}.(\hat{u}\times\vec{v})=1$. Which of the following statement(s) is(are) correct?

(A) There is exactly one choice for such $\vec{v}$

(B) There are infinitely many choice for such $\vec{v}$

(C) If $\hat{u}$ lies in the xy-plane then $|u_1|=|u_2|$

(D) If $\hat{u}$ lies in the xz-plane then $2|u_1|=|u_3|$ **[JEE(Advanced)-2016, 4(–2)]**

VT0182

9. Let O be the origin and let PQR be an arbitrary triangle. The point S is such that $\overrightarrow{OP}.\overrightarrow{OQ}+\overrightarrow{OR}.\overrightarrow{OS}=\overrightarrow{OR}.\overrightarrow{OP}+\overrightarrow{OQ}.\overrightarrow{OS}=\overrightarrow{OQ}.\overrightarrow{OR}+\overrightarrow{OP}.\overrightarrow{OS}$. Then the triangle PQR has S as its

[JEE(Advanced)-2017]

(A) incentre (B) orthocenter (C) circumcentre (D) centroid

VT0183

PARAGRAPH :

Let O be the origin, and $\overrightarrow{OX},\overrightarrow{OY},\overrightarrow{OZ}$ be three unit vectors in the directions of the sides $\overrightarrow{QR},\overrightarrow{RP},\overrightarrow{PQ}$, respectively, of a triangle PQR. **[JEE(Advanced)-2017]**

10. $\left|\overrightarrow{OX}\times\overrightarrow{OY}\right|=$

(A) sin(Q + R) (B) sin(P + R) (C) sin 2R (D) sin(P + Q)

VT0184

11. If the triangle PQR varies, then the minimum value of cos(P + Q) + cos(Q + R) + cos(R + P) is

(A) $\frac{3}{2}$ (B) $-\frac{3}{2}$ (C) $\frac{5}{3}$ (D) $-\frac{5}{3}$

VT0184

12. Let $\vec{a}$ and $\vec{b}$ be two unit vectors such that $\vec{a}.\vec{b}=0$. For some x, y ∈ $\mathbb{R}$, let $\vec{c}=x\vec{a}+y\vec{b}+(\vec{a}\times\vec{b})$. If $|\vec{c}|=2$ and the vector $\vec{c}$ is inclined at the same angle α to both $\vec{a}$ and $\vec{b}$, then the value of $8\cos^2\alpha$ is ______ **[JEE(Advanced)-2018, 3(0)]**

VT0185

13. Three lines

$L_1:\ \vec{r}=\lambda\hat{i},\ \lambda\in\mathbb{R}$,

$L_2:\ \vec{r}=\vec{k}+\mu\hat{j},\ \mu\in\mathbb{R}$ and

$L_3:\ \vec{r}=\hat{i}+\hat{j}+v\hat{k},\ v\in\mathbb{R}$

are given. For which point(s) Q on L_2 can we find a point P on L_1 and a point R on L_3 so that P, Q and R are collinear ? **[JEE(Advanced)-2019, 4(–1)]**

(1) $\hat{k}+\hat{j}$ (2) $\hat{k}$ (3) $\hat{k}+\frac{1}{2}\hat{j}$ (4) $\hat{k}-\frac{1}{2}\hat{j}$

VT0186

14. Let $\vec{a}=2\hat{i}+\hat{j}-\hat{k}$ and $\vec{b}=\hat{i}+2\hat{j}+\hat{k}$ be two vectors. Consider a vector $\vec{c}=\alpha\vec{a}+\beta\vec{b}$, $\alpha,\beta\in\mathbb{R}$. If the projection of $\vec{c}$ on the vector $(\vec{a}+\vec{b})$ is $3\sqrt{2}$, then the minimum value of $\left(\vec{c}-(\vec{a}\times\vec{b})\right).\vec{c}$ equals

[JEE(Advanced)-2019, 3(0)]

VT0187

15. In a triangle PQR, let $\vec{a} = \overline{QR}, \vec{b} = \overline{RP}$ and $\vec{c} = \overline{PQ}$. If $|\vec{a}| = 3$, $|\vec{b}| = 4$ and $\dfrac{\vec{a}.(\vec{c}-\vec{b})}{\vec{c}.(\vec{a}-\vec{b})} = \dfrac{|\vec{a}|}{|\vec{a}|+|\vec{b}|}$,

then the value of $|\vec{a}\times\vec{b}|^2$ is ______ **[JEE(Advanced)-2020]**

VT0188

16. Let a and b be positive real numbers. Suppose $\overrightarrow{PQ} = a\hat{i} + b\hat{j}$ and $\overrightarrow{PS} = a\hat{i} - b\hat{j}$ are adjacent sides of a parallelogram PQRS. Let $\vec{u}$ and $\vec{v}$ be the projection vectors of $\vec{w} = \hat{i} + \hat{j}$ along $\overrightarrow{PQ}$ and $\overrightarrow{PS}$, respectively. If $|\vec{u}| + |\vec{v}| = |\vec{w}|$ and if the area of the parallelogram PQRS is 8, then which of the following statements is/are TRUE ? **[JEE(Advanced)-2020]**

(A) a + b = 4

(B) a – b = 2

(C) The length of the diagonal PR of the parallelogram PQRS is 4

(D) $\vec{w}$ is an angle bisector of the vectors $\overrightarrow{PQ}$ and $\overrightarrow{PS}$ **VT0189**

17. Let $\vec{u}, \vec{v}$ and $\vec{w}$ be vectors in three-dimensional space, where $\vec{u}$ and $\vec{v}$ are unit vectors which are not perpendicular to each other and $\vec{u}.\vec{w} = 1,\ \vec{v}.\vec{w} = 1,\ \vec{w}.\vec{w} = 4$.

If the volume of the parallelopiped, whose adjacent sides are represented by the vectors $\vec{u}, \vec{v}$ and $\vec{w}$, is $\sqrt{2}$, then the value of $|3\vec{u} + 5\vec{v}|$ is____. **[JEE(Advanced)-2021]**

VT0190

18. Let O be the origin and $\overrightarrow{OA} = 2\hat{i} + 2\hat{j} + \hat{k},\ \overrightarrow{OB} = \hat{i} - 2\hat{j} + 2\hat{k}$ and $\overrightarrow{OC} = \frac{1}{2}\left(\overrightarrow{OB} - \lambda\overrightarrow{OA}\right)$ for some $1 > 0$. If

$|\overrightarrow{OB}\times\overrightarrow{OC}| = \dfrac{9}{2}$, then which of the following statements is (are) **TRUE** ? **VT0191**

(A) Projection of $\overrightarrow{OC}$ on $\overrightarrow{OA}$ is $-\dfrac{3}{2}$ **[JEE(Advanced)-2021]**

(B) Area of the triangle OAB is $\dfrac{9}{2}$

(C) Area of the triangle ABC is $\dfrac{9}{2}$

(D) The acute angle between the diagonals of the parallelogram with adjacent sides $\overrightarrow{OA}$ and $\overrightarrow{OC}$ is $\dfrac{\pi}{3}$

19. Let $\hat{i}, \hat{j}$ and $\hat{k}$ be the unit vectors along the three positive coordinate axes. Let

$\vec{a} = 3\hat{i} + \hat{j} - \hat{k}$,

$\vec{b} = \hat{i} + b_2\hat{j} + b_3\hat{k}$, $\quad b_2, b_3 \in R$,

$\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$, $\quad c_1, c_2, c_3 \in R$

be three vectors such that $b_2b_3 > 0$, $\vec{a} \cdot \vec{b} = 0$ and

$$\begin{pmatrix} 0 & -c_3 & c_2 \\ c_3 & 0 & -c_1 \\ -c_2 & c_1 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ b_2 \\ b_3 \end{pmatrix} = \begin{pmatrix} 3-c_1 \\ 1-c_2 \\ -1-c_3 \end{pmatrix}.$$

Then, which of the following is/are TRUE ? **[JEE(Advanced)-2022]**

(A) $\vec{a} \cdot \vec{c} = 0$ (B) $\vec{b} \cdot \vec{c} = 0$ (C) $|\vec{b}| > \sqrt{10}$ (D) $|\vec{c}| \le \sqrt{11}$

VT0215

ANSWER KEY

Do yourself - 1

1. 7 **2** A **3.** $\alpha = 2n\pi \pm \frac{\pi}{3}$ and $\theta = 2n\pi + \frac{\pi}{6}$

Do yourself - 2

4. C **5.** C **6.** A

Do yourself - 3

1. **(a)** $\vec{a}, \vec{d}; \vec{b}, \vec{x}, \vec{z}; \vec{c}, \vec{y}$ **(b)** $\vec{b}, \vec{x}; \vec{a}, \vec{d}; \vec{c}, \vec{y}$ **(c)** $\vec{a}, \vec{y}, \vec{z}$ **(d)** $\vec{b}, \vec{z}; \vec{x}, \vec{z}$

2. $\frac{1}{4}$ **3.** A,B,C **4.** A **5.** A,B,C,D

Do yourself - 4

1. $\frac{12\vec{a} - 13\vec{b}}{5}, -5\vec{b}$ **2.** $\frac{3}{7}\hat{i} - \frac{6}{7}\hat{j} + \frac{2}{7}\hat{k}$ **3.** B **4.** A

5. B **6.** D **7.** D **8.** B **9.** B

10. C **11.** $\vec{a} + \vec{b} + \vec{c} = \vec{0}$

Do yourself - 5

2. B **3.** A **4.** B **5.** B **6.** B

7. $\vec{r} = (\hat{i} - 2\hat{j} + 3\hat{k}) + \lambda(3\hat{i} - 2\hat{j} + 6\hat{k})$

Do yourself - 6

1. $\frac{\pi}{6}$ **2.** –15 **3.** 1

4. $\frac{2}{7}, \frac{2}{49}(3\hat{i} + 6\hat{j} + 2\hat{k})$ and $\frac{190\hat{i} - 110\hat{j} + 45\hat{k}}{49}$ **5.** $-\frac{2}{21}\hat{i} + \frac{32}{21}\hat{j} - \frac{8}{21}\hat{k}$

6. C **7.** B **8.** D **9.** C **10.** D

11. D **12.** D **13.** D **14.** A **15.** A

16. C

Do yourself - 7

1. $\vec{r} = \frac{2}{3}\left(\hat{i}+\hat{j}-2\hat{k}\right)$ **2.** D **3.** **(a)** C; **(b)** A,D

5. $w = -v_1 + 4v_2 - 2v_3$

Do yourself - 8

2. $-\hat{i}-\hat{j}+2\hat{k}$ **4.** B **5.** C **6.** C **7.** A

8. (A) T; (B) U ; (C) P; (D) R ; (E) Q; (F) S; (G) W; (H) V **9.** C

Do yourself - 9

1. $\frac{1}{\sqrt{6}}$ **2.** $\sqrt{\frac{1}{14}}$ **3.** $\frac{3\sqrt{2}}{2}$ **4.** A **5.** B,C,D

Do yourself - 10

1. ±1 **2.** 0 **3.** 7 **4.** Right handed system

5. D **6.** C **7.** C **8.** C

Do yourself - 11

1. B **2.** B **3.** C **4.** B

Do yourself - 12

1. $62,\ 92\hat{i}+102\hat{j}+32\hat{k}$ **2.** A **3.** B **4.** C

5. A,B,D **6.** B,D

Do yourself - 13

1. 3 **2.** linearly dependent. **3.** D **4.** D

5. D **6.** D **7.** B,C,D

Do yourself - 14

1. B **2.** 4 **3.** 1 **4.** B **5.** A

6. B **7.** C **8.** 9

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	B	B	C	A	(i) D, (ii) B, (iii) B	C	B	D	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A	A	A	C	B	A	C	C	D	D
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	C	B	A	D	C	A	D	A	C	A

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,C	A,B,D	B,C	A,C,D	A,C,D	B,C	C,D	A,B,D	A,B,C,D	A,D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A,C	A,B,D	B,C,D	A,B,C,D	A,B,C	B,D	A,B,C	A,C	B,D	A,B,C,D
Que.	21	22	23	24	25					
Ans.	B,D	A,B	A,C	B,D	A,B,D					

EXERCISE # O-3

Que.	1	2	3	4	5	6	7	8	9
Ans.	C	B	C	B	A	2.41 or 2.42	2.44 or 2.45	0.00	B
Q.10	A	B	C	D					
	R	Q,S	Q,R	P,Q,R,S					

EXERCISE # O-4

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	0	5	3	7	3.80	1.04 or 1.05	0.45 or 0.46	5.00	4.00

EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	1	1	4	3	1	3	4	1	2
Que.	**11**	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
Ans.	3	1	1	3	1	30	1.00	8.00	2	4
Que.	**21**	**22**	**23**	**24**						
Ans.	2	36	C	A						

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	5	C	A,B,C	4	A,C,D	Bonus	B,C	B	D
Que.	**11**	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	
Ans.	B	3	3,4	18.00	108.00	A,C	7	A,B,C	B,C,D	